Theory of Automata

Assignment No: 2

Due Date: 21st Oct 2025

Submit hardcopy in Class, and upload soft form on GCR before 5:00 pm on 21 Oct , 2025 (Students, who don’t have class on 21st Oct, should submit it in office on same date)

All languages are defined over Σ = {a, b}

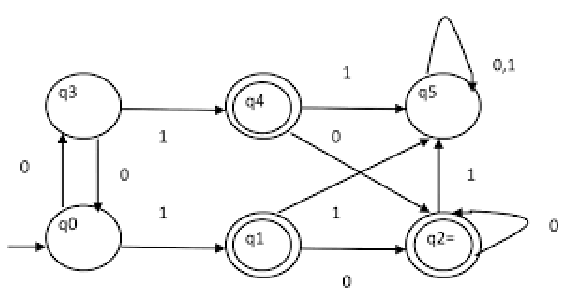
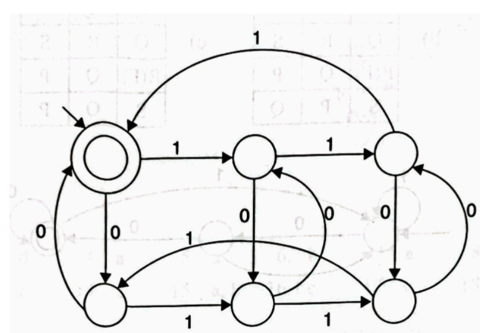
Question No:1 - Kleene’s Theorem Conversions

A. Convert the following a) RE to e-NFA b) e-NFA to DFA c) NFA to DFA d) DFA to minimized DFA

   a. a*
   b. a(ba+a)*
   c. a+b+Δ
   d. (ab+aa)*

Question No:2 - Kleene’s Theorem

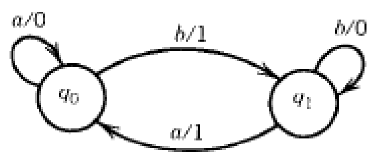
Consider the DFAs given. Use Kleene’s theorem to solve it.

DFA 1	DFA 2
	

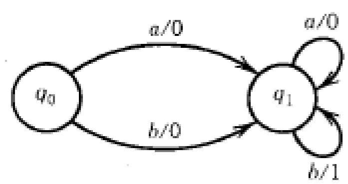
a. DFA-1 + DFA-2 (+ representing Union)
b. DFA-1.DFA-2 (. representing Concatenation)
c. (DFA-1)*

Question No:2 - <u>Mealy and Moore Machines</u>

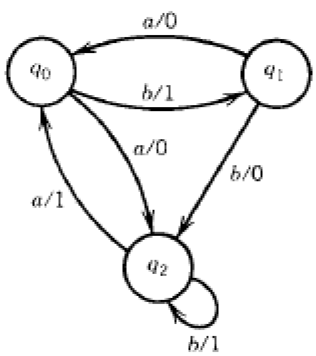
B. Construct a Moore machine that subtracts 1 from a binary number. Assume the input number is always greater or equal to 1.
C. Construct a Mealy machine that adds 2 to a binary number.
D. Convert the following to Moore machines:



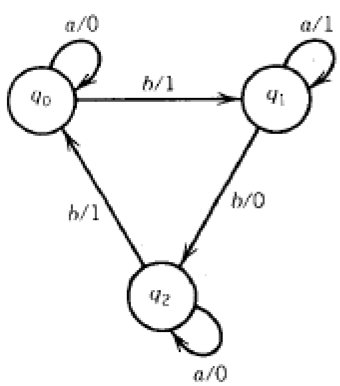
a.



b.



c.



d.